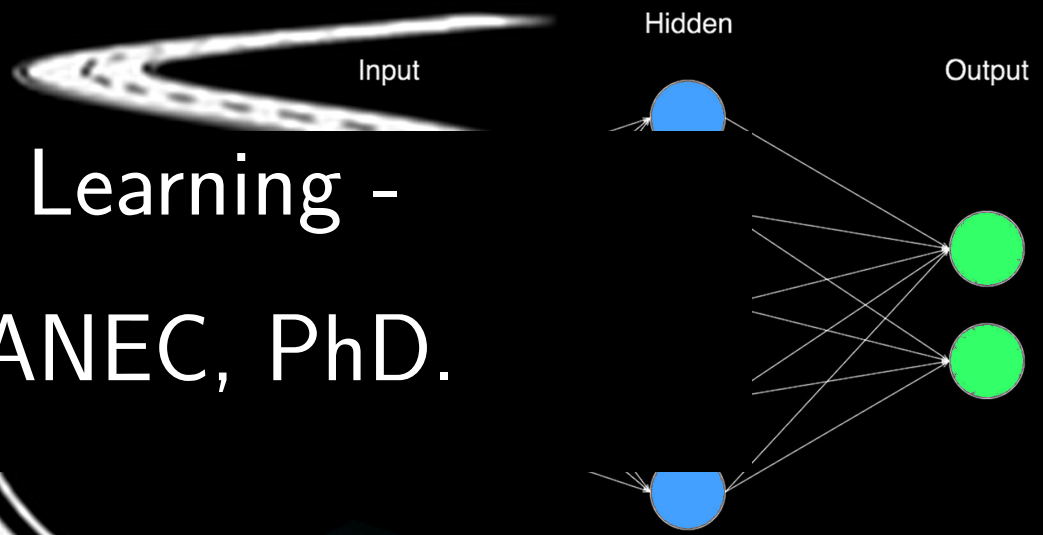
$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha\left[r_{t+1} + \lambda \max_a Q(s_{t+1}, a) - Q(s_t, a_t)\right]$$

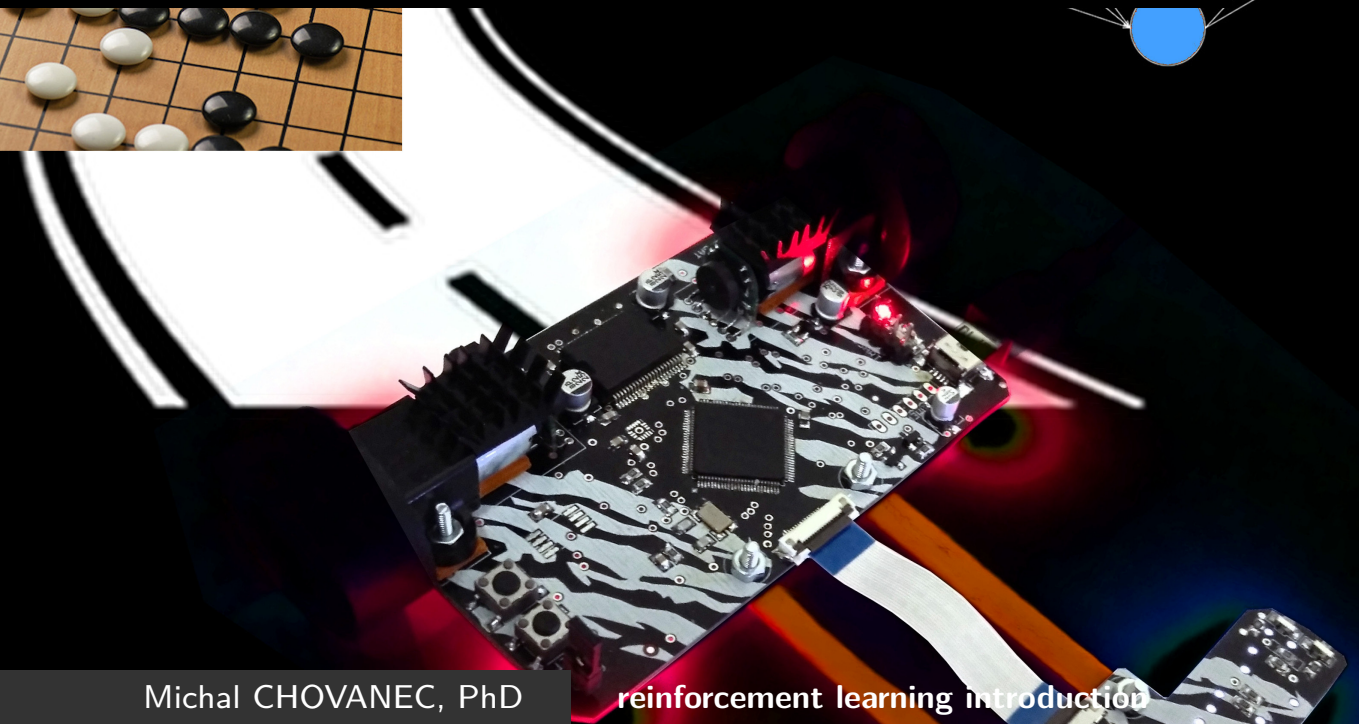
(The New Action Value = The Old Value) + The Learning Rate × (The New Information − the Old Information)



Reinforcement Learning - Introduction

Michal CHOVANEC, PhD.





Overview

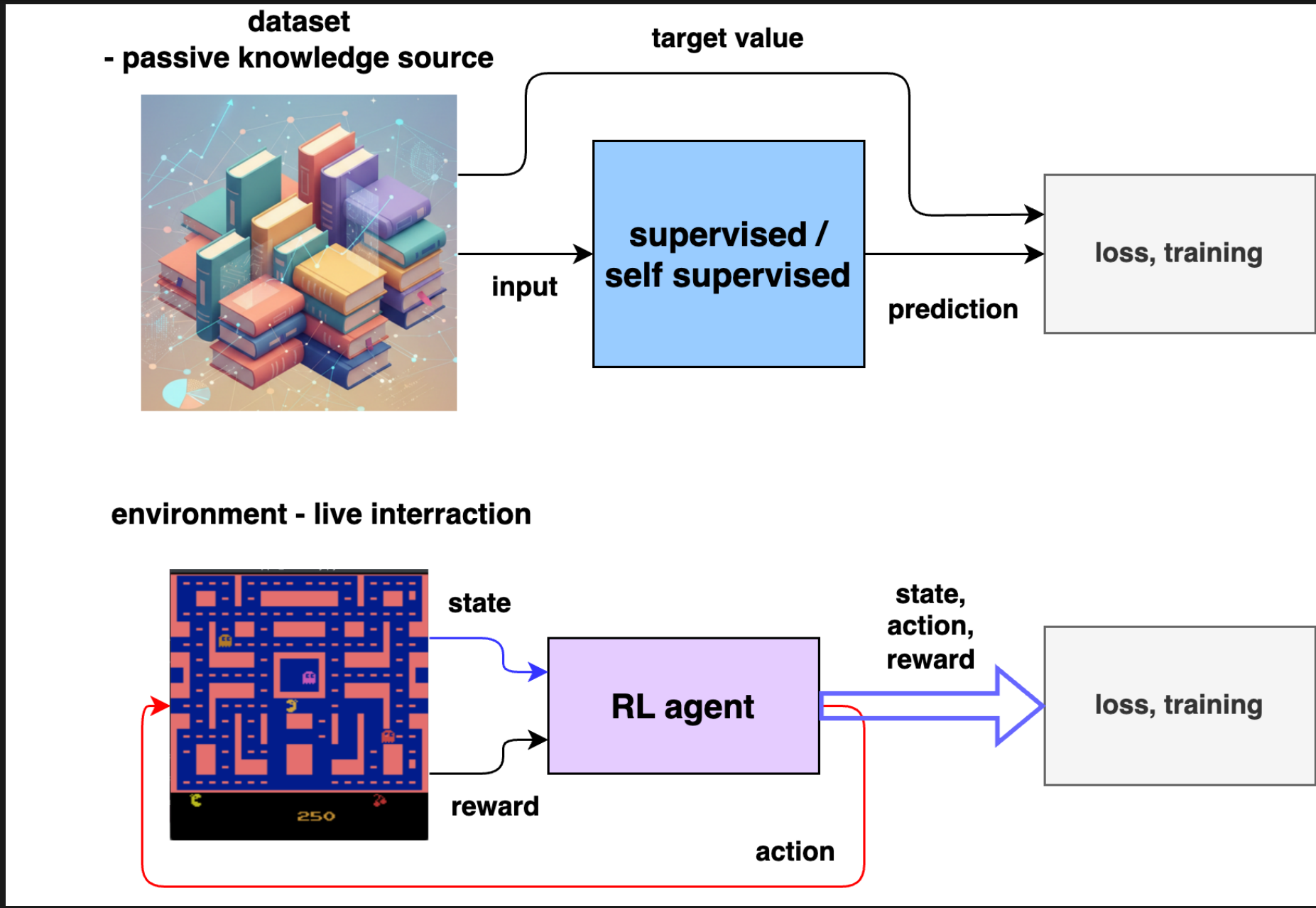
1. Agents, Principle
2. Policy Gradients, Advantage-Actor-Critic
3. Practical Example - Lunar Lander
4. Cookbook and Literature

Reinforcement learning

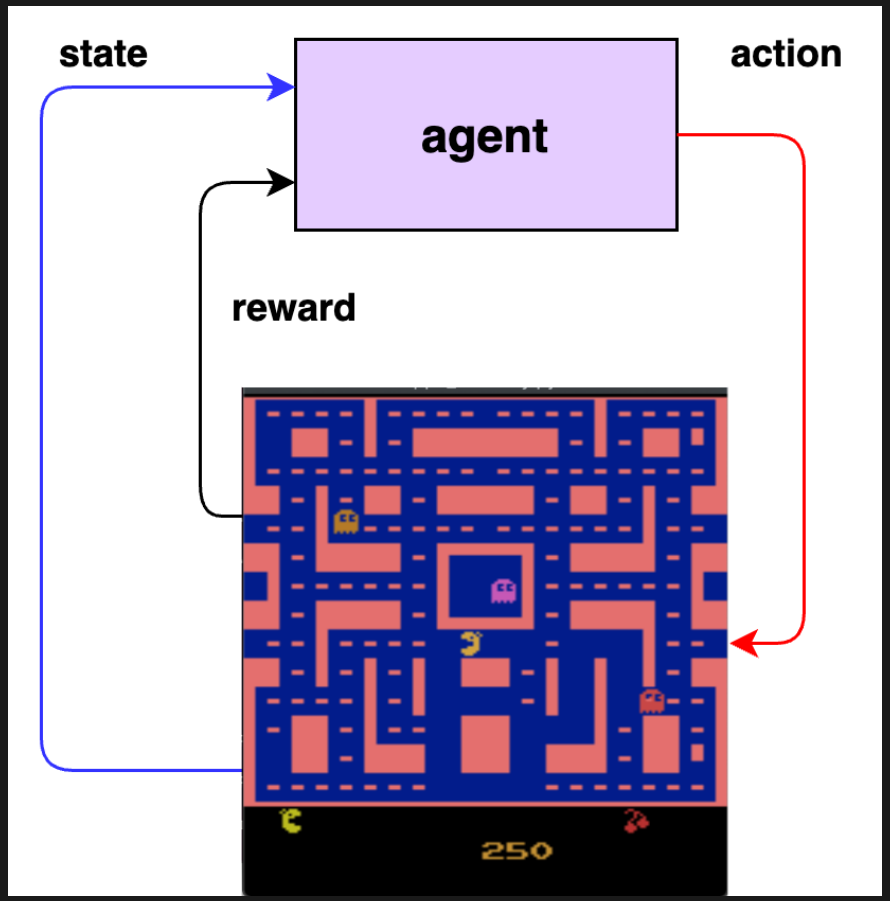




Reinforcement learning - principle



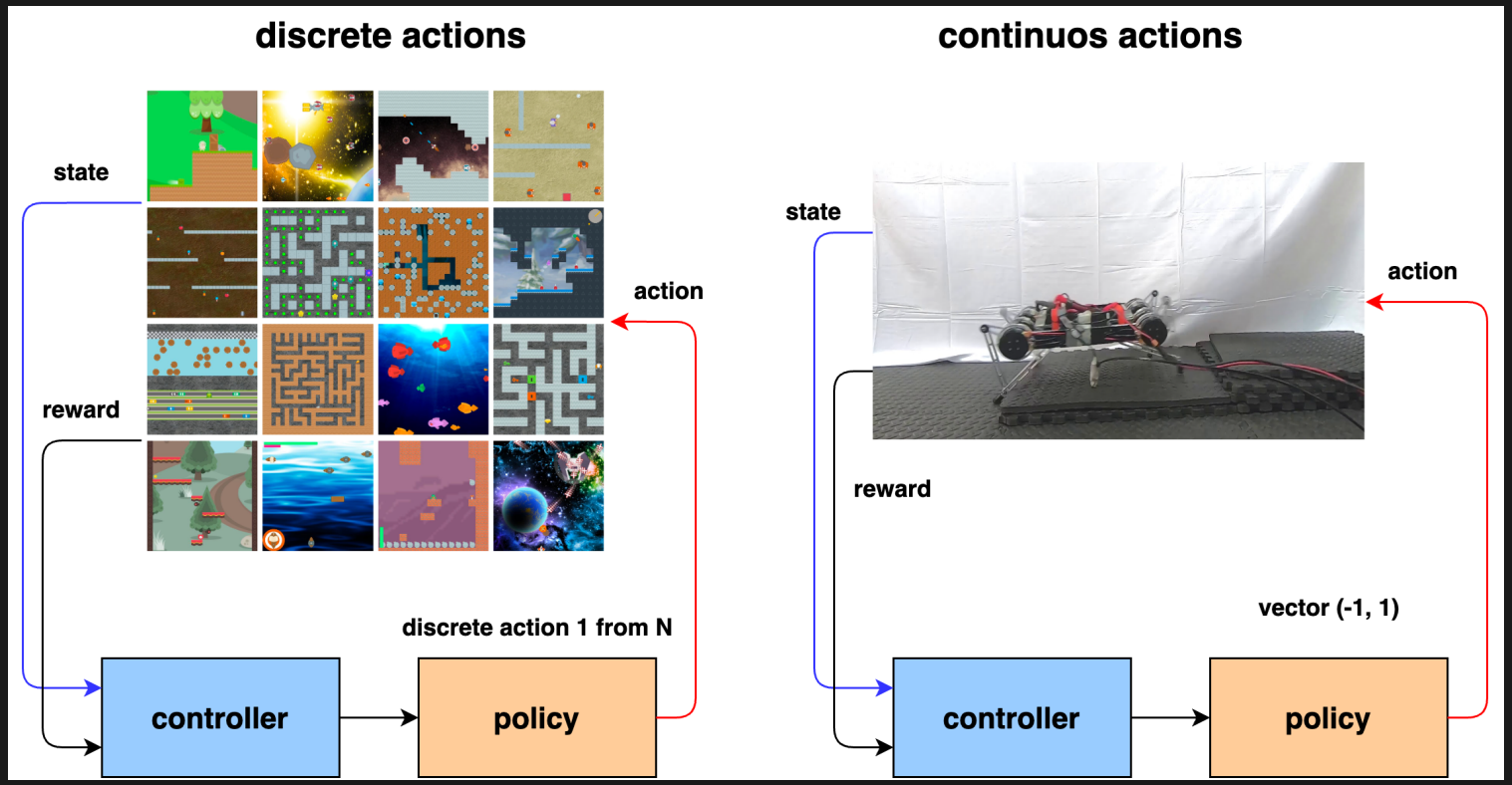
Reinforcement learning - principle



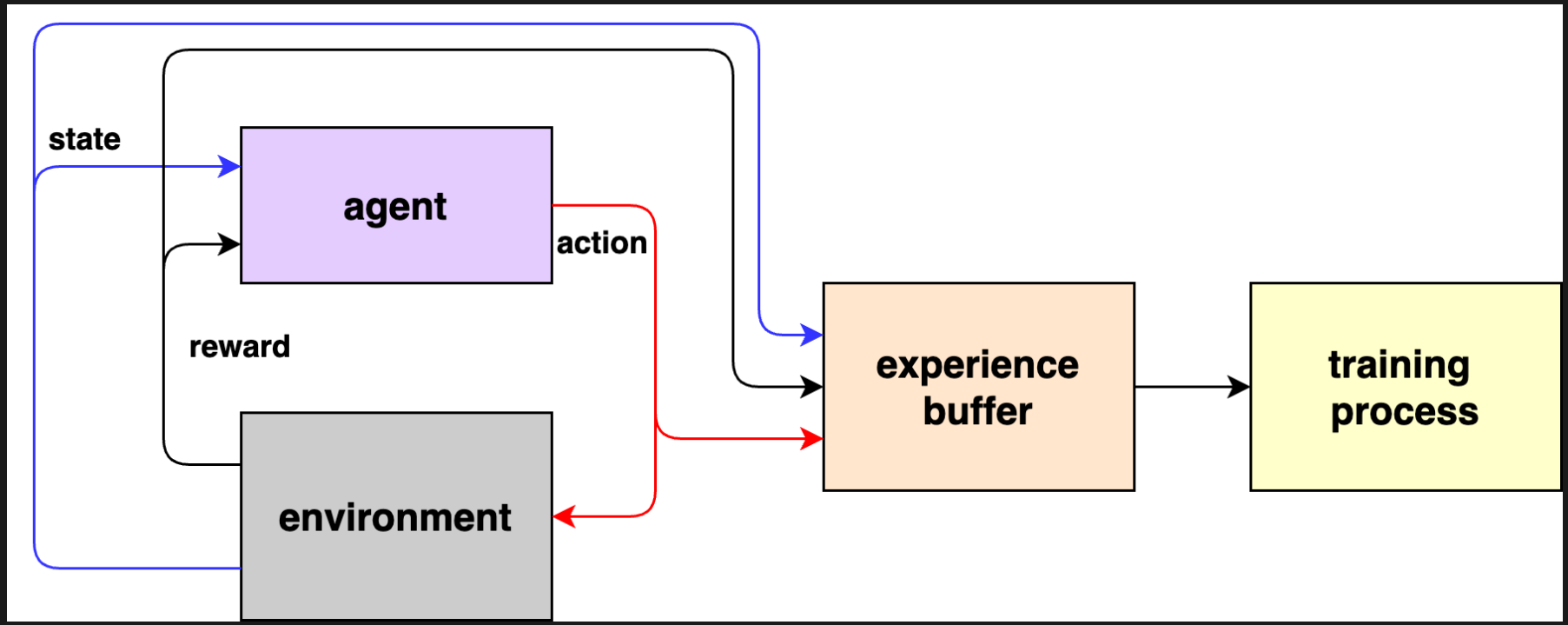
- **active environment interraction**
- obtain state
- select action
- exectute action
- learn from experiences

Action space

- discrete action space
  - keys, keypad
- continuous action space
  - motors, PWMs, steering, torque control

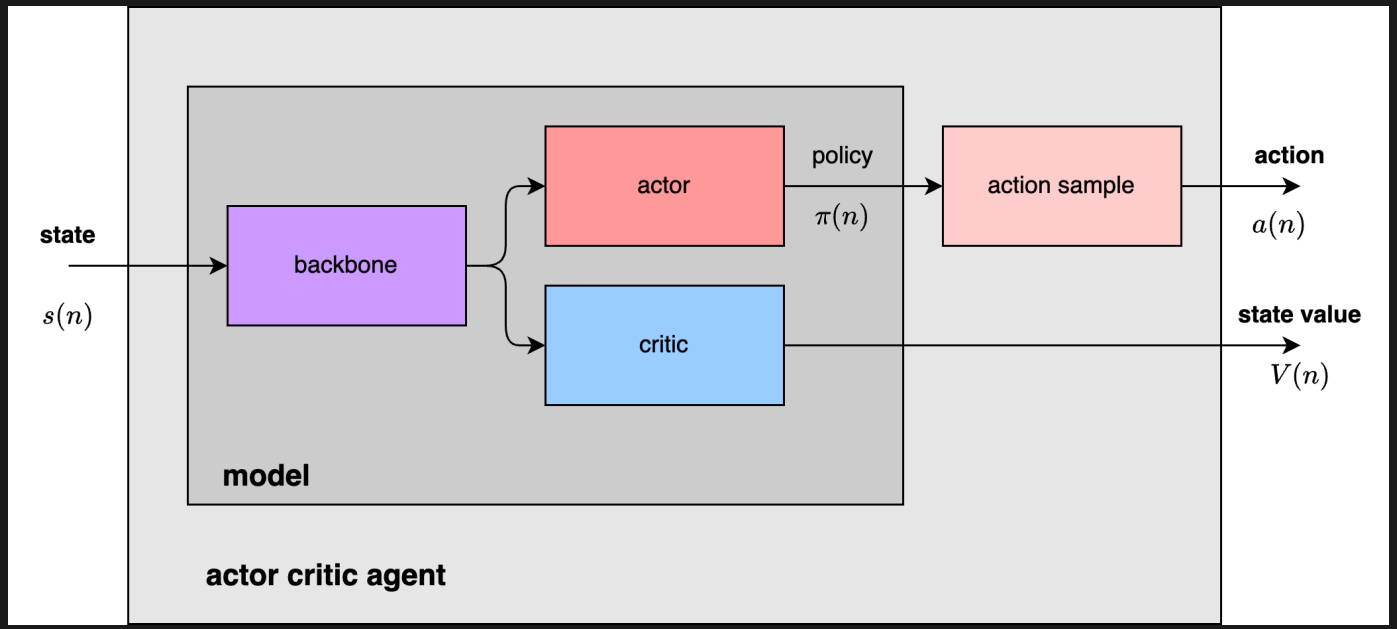


reinforcement learning - training

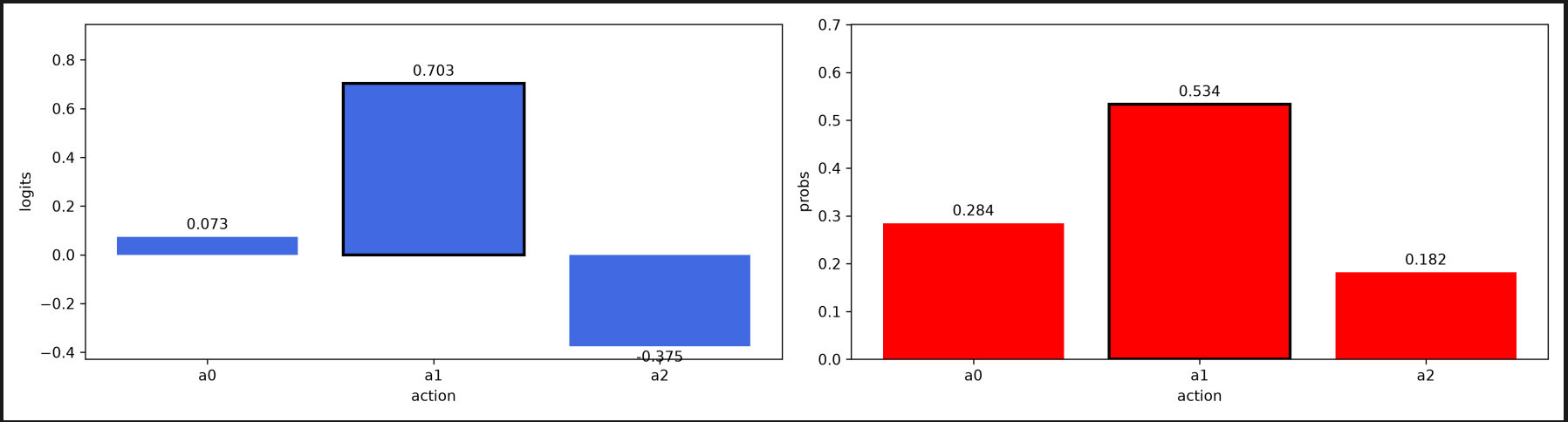


- obtain state
- select action
- exectute action
- learn from experiences

Policy Gradients, Advantage-Actor-Critic (A2C)



Action sampling



1. logits :

$$l(s) = f_\theta(s)$$

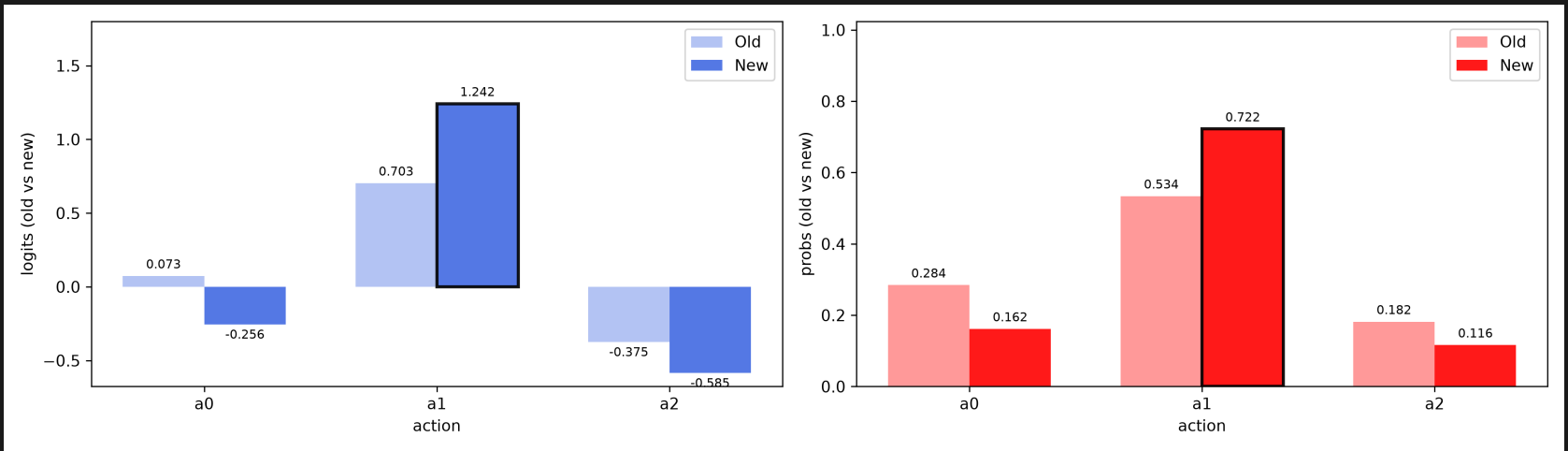
2. probs :

$$\pi_\theta(a|s) = softmax(l(s))$$

3. action sampling :

$$a \sim \pi_\theta(.|s)$$

Actor training



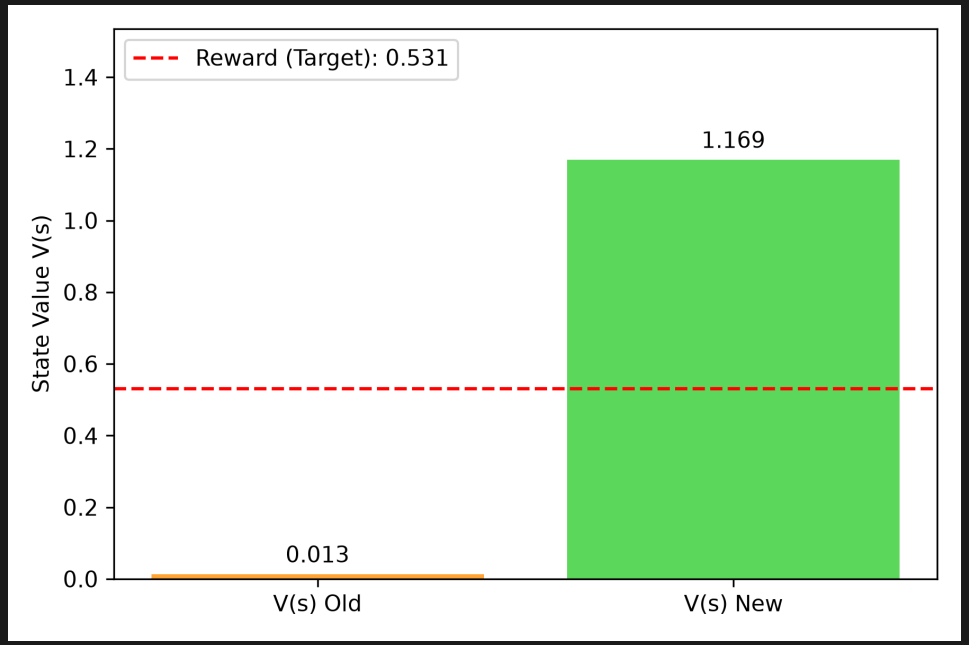
1. advantages (simplified)

$$A(s,a) = reward - V(s)$$

2. policy loss

$$\mathcal{L}_\theta(s,a) = -A(s,a) log \pi_\theta(a|s)$$

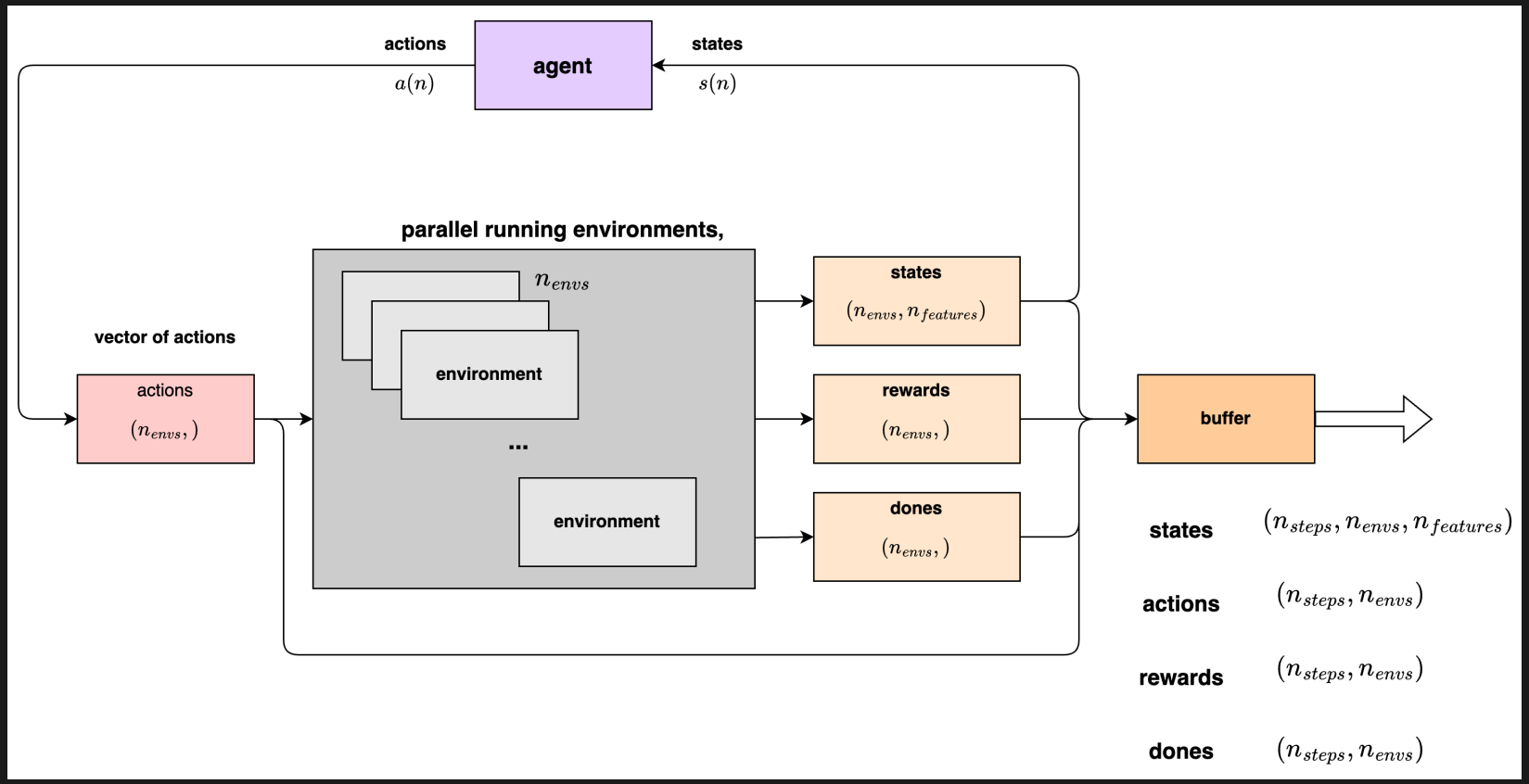
Critic training



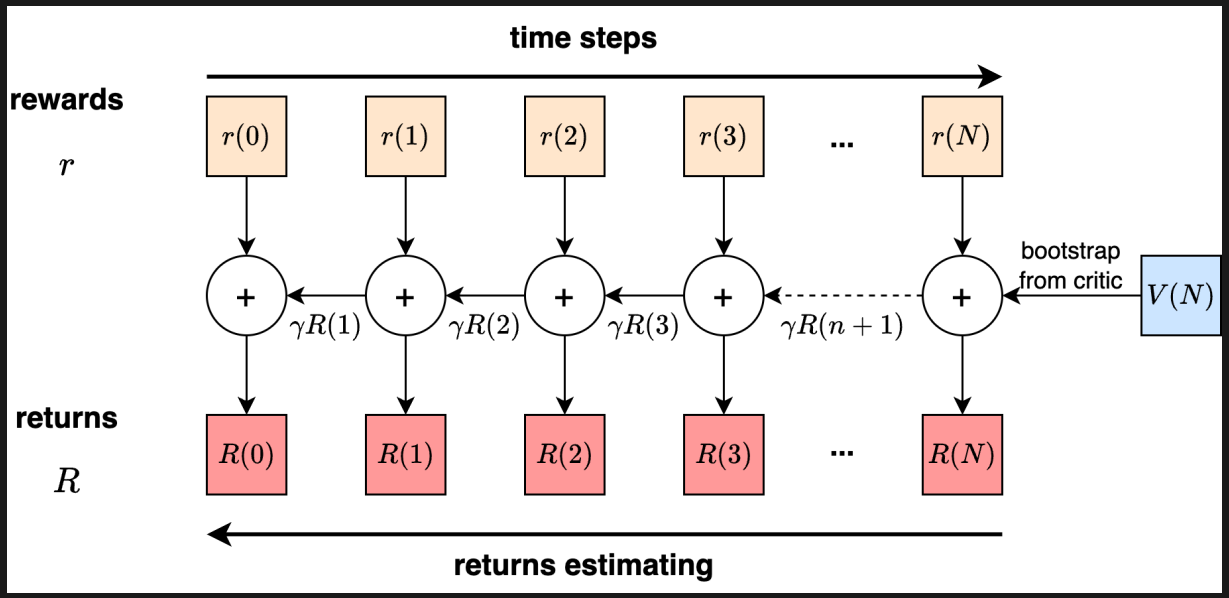
1. critic loss

$$\mathcal{L}_\Phi(s) = (V(s) - r)^2$$

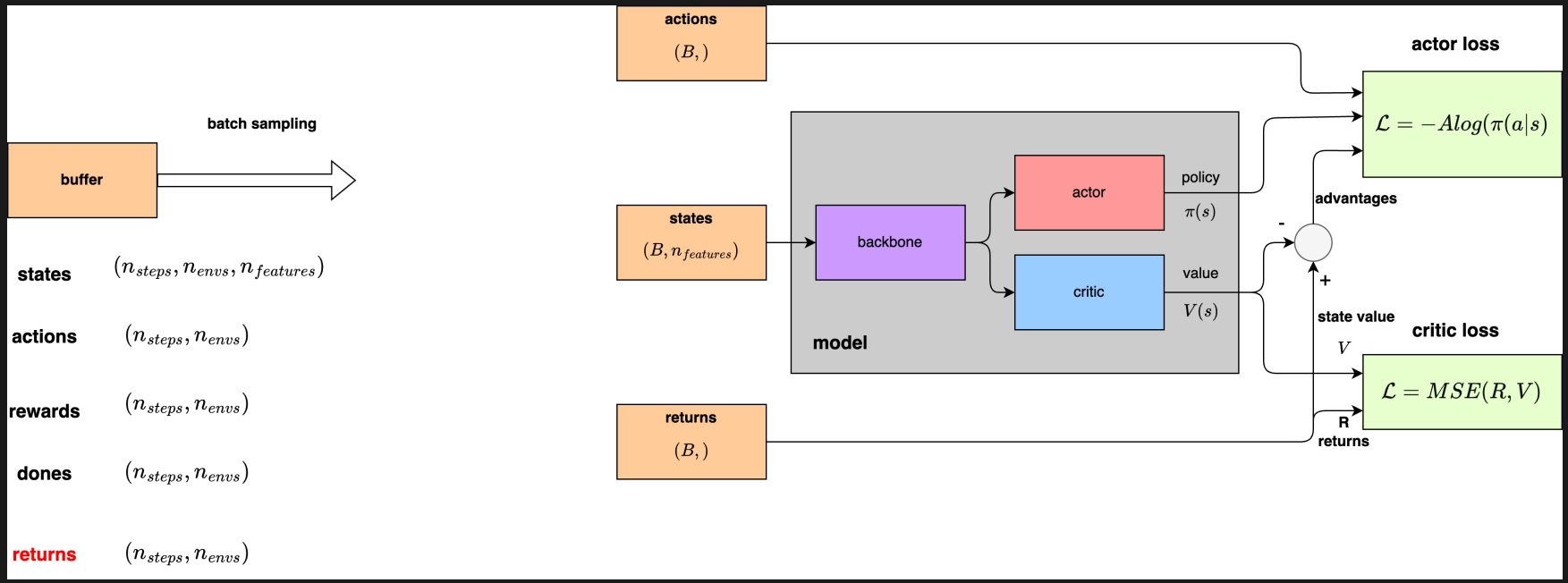
A2C training



A2C training



A2C training



deep Q learning

$$Q(s, a; \theta) = \underbrace{R}_{\textit{reward}} + \underbrace{\gamma \max_{a'} Q(s', a'; \theta^-)}_{\textit{discounted future reward}}$$

$$\mathcal{L}(\theta) = \left(R + \gamma \max_{a'} Q(s', a'; \theta^-) - Q(s, a; \theta) \right)^2$$

Q&A

michal.nand@gmail.com
https://github.com/michalnand/lectures/